

Library Book Request

Capstone Project Document

1. INTRODUCTION

1.1 Project Overview

The **Library Book Request System** is a ServiceNow application developed to automate the process of requesting books from the college library. Students can submit book requests online by entering the required details. The system validates the issue date, generates a unique request number automatically, stores the request in a custom table, and sends an email notification to the librarian.

1.2 Purpose

The purpose of this project is to replace the manual library book request process with an automated system that improves efficiency, reduces paperwork, and enables better tracking of requests.

2. IDEATION PHASE

2.1 Problem Statement

The existing library book request process is manual, making it difficult to track requests and causing delays in processing. There is also a possibility of incorrect issue dates and missing request details.

2.2 Empathy Map Canvas

Think

- I need the required book quickly.
- I want an easy request process.

Feel

- Frustrated with manual paperwork.
- Wants quick confirmation.

Say

- I need to know my request status.

- The request process should be simple.

Do

- Fill out the online request form.
- Submit the request.

2.3 Brainstorming

Possible Solutions:

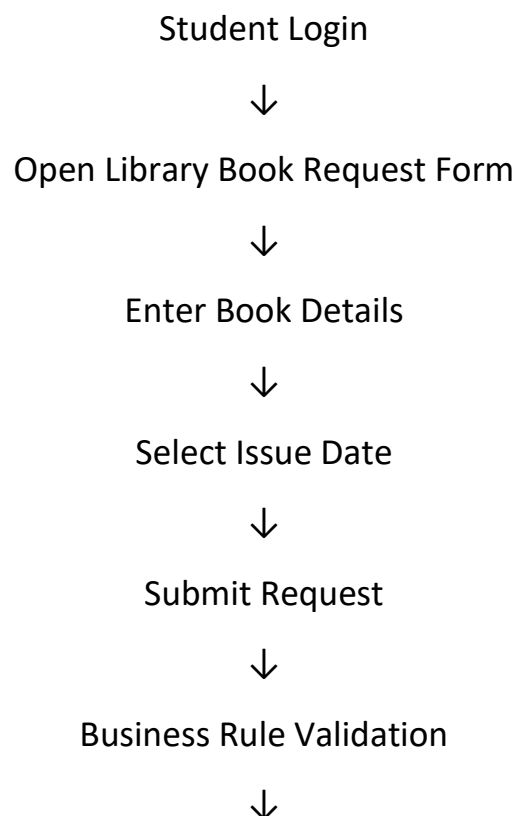
- Manual paper request
- Email request
- Google Form
- **ServiceNow Library Book Request Application (Selected Solution)**

Reason:

ServiceNow provides automation, validation, notifications, and request tracking.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map



Auto Request Number Generated



Email Notification to Librarian



Request Completed

3.2 Solution Requirement

Functional Requirements

- Student can request books.
- Issue date should not be in the past.
- Request number should be generated automatically.
- Librarian should receive an email notification.
- Request should be stored in the custom table.

Non-Functional Requirements

- User-friendly interface
- Fast response time
- Secure data storage
- High availability

3.3 Data Flow Diagram

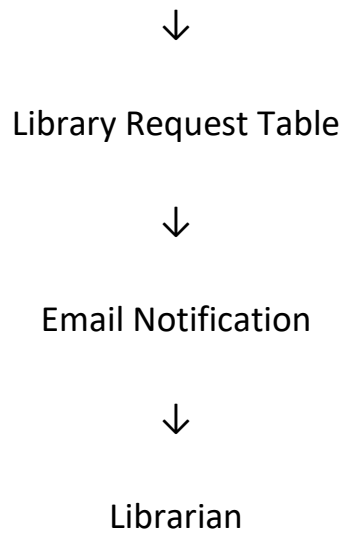
Student



Library Book Request Form



Business Rule



3.4 Technology Stack

Component	Technology
Platform	ServiceNow
Database	Custom Table
Validation	Business Rule
Notification	Email Notification
Number Generation	Auto Number

4. PROJECT DESIGN

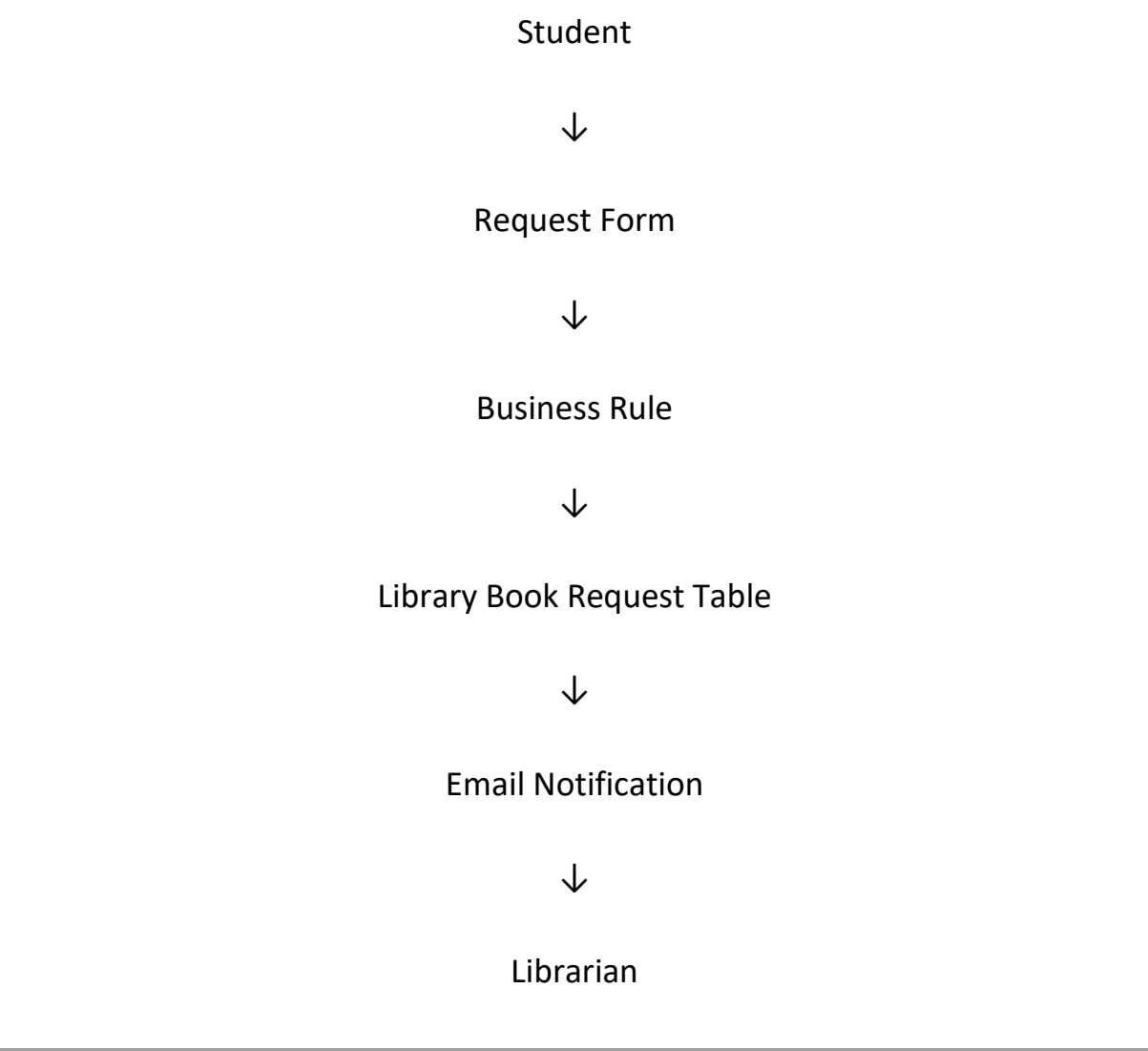
4.1 Problem Solution Fit

The application eliminates manual paperwork by providing an automated online request system. It validates user input, generates unique request numbers, and sends notifications to the librarian.

4.2 Proposed Solution

Develop a ServiceNow application where students can submit book requests. The application validates issue dates, generates request numbers automatically, stores request details, and notifies the librarian via email.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

Activity	Status
Requirement Analysis	Completed
Custom Table Creation	Completed
Form Design	Completed
Business Rule Development	Completed

Activity	Status
Auto Number Configuration	Completed
Email Notification	Completed
Testing	Completed

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Parameter	Result
Request Submission Time	Less than 2 seconds
Issue Date Validation	100% Accurate
Auto Number Generation	Less than 1 second
Email Notification	Successfully Delivered
Overall Performance	Excellent

7. RESULTS

The Library Book Request application successfully:

- Created library book requests.
- Generated unique request numbers.
- Validated issue dates.
- Sent email notifications to the librarian.
- Stored request details successfully.

(Include screenshots of the form, business rule validation, generated request number, request list, and email notification.)

8. ADVANTAGES & DISADVANTAGES

Advantages

- Easy online request submission.
- Eliminates paperwork.
- Automatic request number generation.
- Validates issue dates.
- Email notifications improve communication.
- Easy tracking of requests.

Disadvantages

- Requires internet connectivity.
- Depends on the ServiceNow platform.
- Only authorized users can access the application.

9. CONCLUSION

The **Library Book Request System** successfully automates the library request process using ServiceNow. The project ensures accurate request validation, automatic request number generation, and timely notifications to the librarian. This improves efficiency, reduces manual work, and enhances the user experience.

10. FUTURE SCOPE

- Book availability checking.
- Online approval workflow.
- SMS notifications.
- Mobile application support.
- Barcode/QR code integration.
- Student dashboard for tracking requests.
- Report generation and analytics.

